TASK 3: ENGINE, FUEL AND COOLING SYSTEM DIAGNOSIS

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 6 | v1.1

TIME
90 minutes (adapt to workplace conditions)

EVIDENCE
Practical | Knowledge | Documentation

PROGRAMME POINT
Weeks 75-90 | Months 20-24

Your task

Investigate a heavy vehicle with reduced performance or an abnormal temperature/pressure concern. Use technical data, live information and mechanical tests to locate the cause.

What you must do

1. Confirm symptoms, operating conditions, fluid levels and fault history.
2. Inspect air, fuel, lubrication and cooling systems before intrusive testing.
3. Use scan data and authorised pressure, temperature or flow tests.
4. Compare results, identify interactions and isolate the likely root cause.
5. Record repair action, environmental controls and verification requirements.

What good evidence looks like

SAFE CONTROL

Allow systems to cool and depressurise as specified; do not open high-pressure fuel systems without the manufacturer procedure and suitable authority.

CORRECT METHOD

A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE

Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS

A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle or engine rig
- Manufacturer test data
- Scan tool and pressure/temperature equipment
- PPE, spill control and job card

SOW ALIGNMENT

Block 6 - Engines, Fuel, Emissions, Thermal Management and Emerging Technology

KSB evidence focus: S2, S4, B3; reinforced K1-K4, K6, K10, S1, S3, S5-S6, S8-S10, B2

Task reference: HV_L3_Block_6_Task_3_Engine_Fuel_and_Cooling_System_Diagnosis

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Coolant temperature / pressure		Cooling specification	
Fuel supply pressure		Fuel specification	
Boost / air-flow result		Boost specification	
Oil pressure / condition		Diagnosis / action	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why are basic fluid and visual checks completed before advanced tests?

2. How can an air or cooling fault affect engine performance?

Knowledge check - continued

3. Explain controls for hot, pressurised or high-pressure fuel systems.

4. How do you distinguish a sensor value from the true physical condition?

5. What post-repair tests demonstrate a reliable result?

Assessor/Workplace Mentor observation

- [ ] Controls hot, pressurised and contaminated systems.
- [ ] Uses service information to plan meaningful tests.
- [ ] Records scan and physical measurements with conditions.
- [ ] Links interacting engine systems to root cause.
- [ ] Plans/records repair and operational verification.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- [ ] PASS
- [ ] REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

FAULT SCENARIOS AND EVIDENCE

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 6 | Task 3

How to use this sheet

Work through the listed scenarios during the block using a vehicle, safe fault rig or supported case study. Record which scenario was used for this task and reference the actual measurements and assessor observation on the existing practical record. A discussion or case study alone is not evidence of practical competence.

1. Lack of power

Checks: Confirm when derate occurs; check fault history, intake/boost, fuel supply, exhaust restriction and live data.

Evidence: Record measured values, root cause and loaded verification where authorised.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

2. Oil pressure fault

Checks: Confirm warning against manufacturer limits with an approved mechanical test; distinguish sensor/circuit from real low pressure.

Evidence: Record pressure at stated speed/temperature; stop engine if low pressure is confirmed.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

3. Misfire

Checks: Confirm cylinder/event data and operating conditions; compare fuel, air, electrical and mechanical evidence using approved tests.

Evidence: Record cylinder identification, test results, repair/referral and smooth-running check.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

Scenario selected for the original task assessment: _____

For any safety-critical fault, stop or control the work and follow the manufacturer, DVSA and workplace procedures. Use the original task pages for findings, feedback, PASS/REFER and signatures.

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.